MILITARY SPECIFICATION
SEMICONDUCTOR DEVICES
TRANSISTOR, NPN, SILICON
TYPE 2N1493

This amendment forms a part of Military Specification MIL-S-19500/247(EL) 1/, dated 18 June 1962, and is approved by the US Army Electronics Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* Following the title, add:

"[Inactive for new design after the date of amendment 4]"

1.2, P_C column: Delete "3.5" and substitute "3.0". θ_{J-C} column: Delete "50" and substitute "58". Also, footnote 1/, delete and substitute:

"1/ This power dissipation is 1000 hours expected life at T_A = 25° ±3°C, with heat sink employed; relatively, without a heat sink, P_C = 0.5 W. For power ratings at ambient temperatures above 25°C, see figure 2."

PAGE 3

3.5.1: Delete in its entirety.

* Add the following new paragraph:

"3.6 Manufacturer eligibility. To be eligible to supply transistors to this specification, the manufacturer shall have a product assurance program approval in accordance with MIL-S-19500 for at least one line."

* 4.2 and 4.2.1, delete and substitute:

"4.2 Qualification and acceptance inspection. Qualification shall not be required. Acceptance inspection shall be in accordance with MIL-S-19500 and as specified herein. Groups A and B inspection shall consist of the examinations and tests specified in tables I and II respectively, herein.

"4.2.1 Sampling plan. The sampling plan shall be on a 100 percent basis for group A and the LPTD plan for group B."

* 4.2.1.1, 4.2.2, 4.2.3, and 4.2.4: Delete in their entirety.

PAGE 4

* Table I, delete "LTPD" and "Max. Acc. No." columns. Make this same deletion on page 5.

1/ Activity EL changed to ER.

FSC 5961

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Table II, Subgroup 6, Thermal resistance test, Limits, Max column: Delete "50" and substitute "58". Table II, Subgroup 7, High-temperature life, delete and substitute the following:

"1032	High-temperature life (nonoperating) End-points tests Same as for Subgroup 2, above"	T_{stg} = +200°C time = 340 hours 2/	7 ----------

Table II, Subgroup 8, Steady-state operation life, delete existing data and substitute the following:

"1027	Steady-state operation Life	T_C = +100°C with heat sink P_C = 1.5 W V_{CB} = 40 V time = 340 hours 2/	7 ----------
	End-points tests Same as for Subgroup 2, above"		

Following Table II, add footnote 2/ as follows:

"2/ Samples that have been subjected to Group B, 340 hour life test, may be continued on test to 1000 hours in order to satisfy Group C life test requirements. These samples shall be predesignated, and shall remain subjected to the Group C 1000 hour acceptance evaluation after they have passed the Group B 340 hour acceptance criteria. The cumulative total of failures found during 340 hour test and during the subsequent interval up to 1000 hours shall be computed for 1000 hour acceptance criteria."

Add the following Table III, Group C tests:

"Table III. Group C inspection. 1/

Test Method per MIL-STD-750	Examination or test	Conditions	LTPD
1031	Subgroup 1 High Temperature life (nonoperating) End-point tests Same as for Subgroup 2 of Group B	T_{stg} = +200°C 2/	λ = 10
1026	Subgroup 2 Steady-state operation life	T_C = +100°C with heat sink P_C = 1.5 W V_{CB} = 40 V 2/	λ = 10

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End-point test
Same as for subgroup 2
of Group B

1/ Group C inspection shall be conducted on the initial lot and thereafter every six months during production.

2/ Samples that have been subjected to Group B, 340 hour life test, may be continued on test to 1000 hours in order to satisfy Group C life test requirements. These samples shall be predesignated and shall remain subjected to the Group C 1000 hour acceptance evaluation after they have passed the Group B 340 hour acceptance criteria. The cumulative total of failures found during 340 hour test and during the subsequent interval up to 1000 hours shall be computed for 1000 hour acceptance criterial."

PAGE 9

* 6.2: Delete in its entirety.

PAGE 11

FIGURE 2: Delete and substitute new figure 2 as shown on page 4 of this amendment.

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous amendment.

Custodian:
Army - ER

Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5961-A898)

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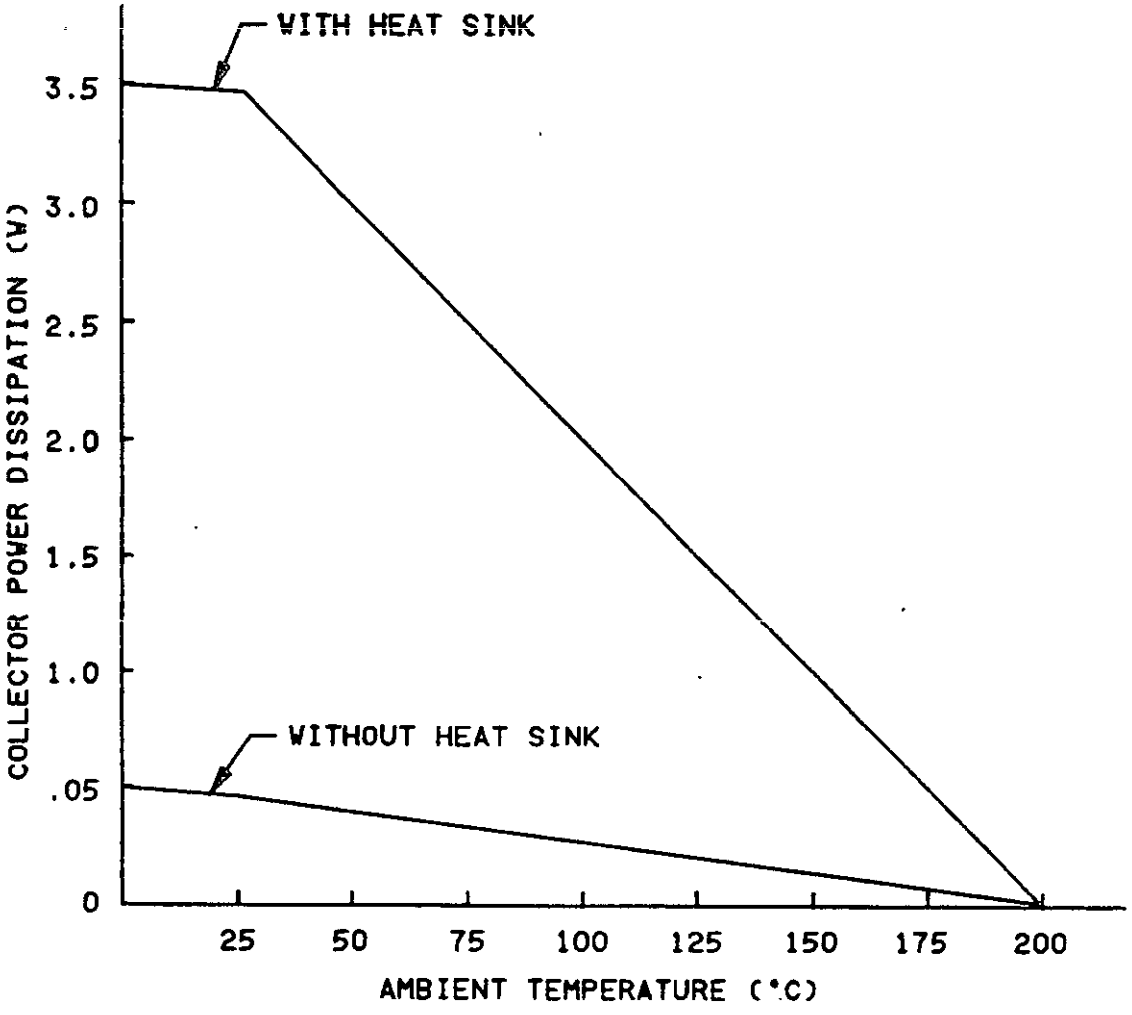


FIGURE 2. Power dissipation derating curve.

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